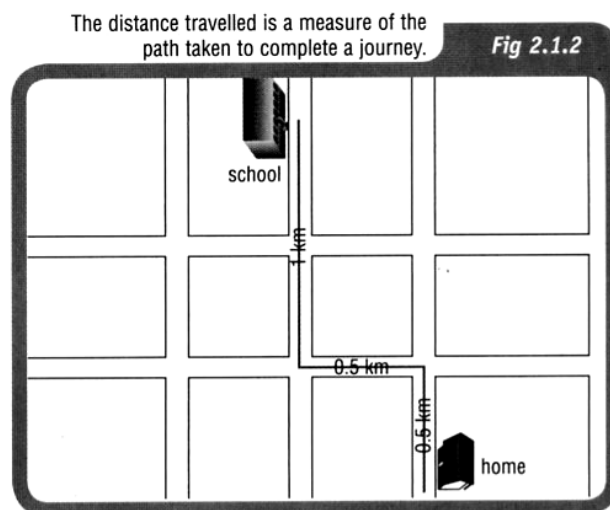


DISTANCE AND DISPLACEMENT

Distance (d): how far a body moves in any direction.

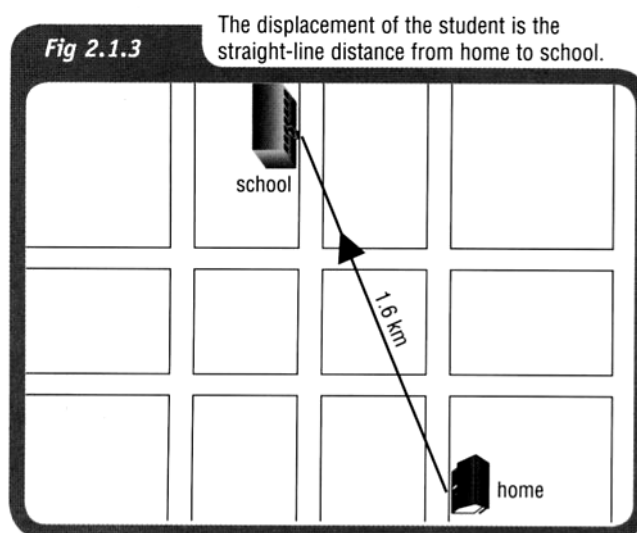
e.g. The diagram below shows the movement of a student between home and school.



The distance travelled is 2.0 km.

Displacement (s): change of position of a body from the start of the motion to the finish of the motion, and is in a particular direction.

e.g. The diagram below shows the displacement of the student when travelling to school.



The straight-line distance is 1.6 km. No direction is indicated in the diagram.

If the distance you walk from home to school is 2 kilometres, the distance you travel from home to school and back home again is 4 kilometres. If you start at home and finish the day's journey at home, your starting and finishing points are the same position (home) then your **displacement for the day is actually 0 km**.

In Physics, distance is known as a *scalar quantity* and displacement is a *vector quantity*.

See the notes on Scalars and Vectors.

SPEED AND VELOCITY

Speed: distance moved per unit time.

$$\text{Speed} = \frac{d}{t}$$

where speed = how fast the object travels for the entire journey (ms^{-1})
 d = distance travelled for the entire journey (m)
 t = time taken for the entire journey (s)

Speed is a *scalar* quantity as direction is *not* important. Velocity is a *vector* quantity and has a *direction* involved.

Velocity: displacement moved in a given direction per unit time.

$$v = \frac{s}{t}$$

where v = velocity in a given direction (ms^{-1})
 s = displacement in a given direction (m)
 t = time taken to move in a given direction (s)

Note

The units for speed and velocity are metres per second (ms^{-1}), not kmh^{-1} .

To convert kmh^{-1} to ms^{-1} , divide the reading in kmh^{-1} by 3.6.

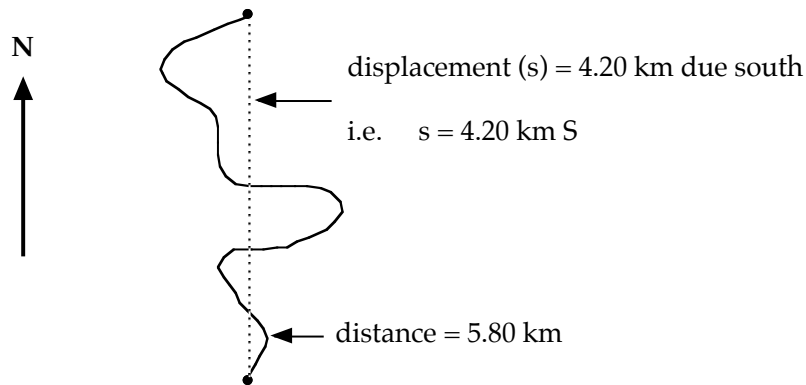
Example 1

$$\begin{aligned} 60.0 \text{ kmh}^{-1} &= \frac{60.0}{3.6} \\ &= 16.7 \text{ ms}^{-1} \end{aligned}$$

Example 2

A cyclist rides along a coast road that meanders and changes direction constantly. He covers a distance of 5.80 km in a time of 12.2 minutes. However the straight line distance between his start and finish points is only 4.20 km in a direction due south. Calculate:

- (a) the average speed of the cyclist.
- (b) his average velocity.



(a)

$$\begin{aligned}\text{Speed} &= \frac{d}{t} \\ &= \frac{5.80 \times 10^3}{7.32 \times 10^2} \\ &= 7.923 \text{ ms}^{-1}\end{aligned}$$

$$\therefore \text{speed} = 7.92 \text{ ms}^{-1}$$

(b)

$$\begin{aligned}v &= \frac{s}{t} \\ &= \frac{4.20 \times 10^3}{7.32 \times 10^2} \\ &= 5.738 \text{ ms}^{-1}\end{aligned}$$

$$\therefore v = 5.74 \text{ ms}^{-1} \text{ South}$$

**The direction is important
- make sure you put it in!**